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1. INTRODUCTION

BK and JC viruses are human polyomaviruses from the family *Polyomaviridae*. In healthy people, the BK virus causes lifetime infection with no symptoms, but in immune weakened patients, such as those who had solid organ transplant or a bone marrow transplant, BK virus may cause serious disease. According to sero epidemiological research, adult seroprevalence is 90% globally (Singh et al., 2009). In immune-compromised individuals, they may induce frequent and severe viruria and viremia. The BK virus is connected to two key post-transplant complications: BK-related nephropathy (BKVN) in kidney transplant receivers & BK-associated haemorrhagic cystitis (BKV-HC) in the recipients with allogeneic hematopoietic stem cell therapy. The degree of immunosuppression and the usage of immunosuppressive medications may influence the lifecycle of BK virus in terms of replication & clinical manifestation and illness in transplanted patients. BK virus causes haemorrhagic cystitis in allogeneic hematopoietic stem cell transplantation (HSCT) patients, resulting in substantial morbidity and a lengthy hospital stay. Allogeneic HSCT, unrelated donor transplants, or myeloablation conditioning, are all risk factors for BKV-HC in HSCT patients. It typically begins around the moment of neutrophil engraftment, especially in the case of allogeneic HSCT, although it may occur anywhere from 2 weeks to 6 months after the procedure (Cesaro et al., 2008). There is currently no standard therapy for BKV haemorrhagic cystitis. Bladder irrigation, blood transfusion, or therapy using hyperbaric oxygen therapy, cidofovir, or fluoroquinolones leflunomides, are examples of supportive measures. (Azzi et al., 1999) JC virus, like the BK virus, reactivates and sheds in urine in HSCT patients (albeit less often than the BK virus). It may confuse the severity of illness with respect to haemorrhagic cystitis in several bone marrow transplantation patients while being less linked with clinical illness (Gardner et al., 1971).

BK reactivation and the prevalence of haemorrhagic cystitis in patients with hematopoietic stem cell transplant (HSCT) are both poorly understood in India. This will be pilot research to see whether BK polyomavirus reactivation can be detected in HSCT patients and if there is a clinical connection in our setting.

2. REVIEW OF LITERATURE

The polyomavirus that belongs to the family *Polyomaviridae* includes BK and JC-virus, BK Virus is common in stem cell transplant recipients and JC in Progressive Multifocal Leukoencephalopathy (PML) patients.(Ambalathingal et al.,2017). BK viruses are prevalent in nature, producing hidden infections in the urinary system that may last for years before reappearing, usually after immune suppression. Its name is derived from the surname of the first patient in whom it was discovered. Polyomaviruses are non-enveloped, double stranded DNA viruses with circular genomes of about 5000 base pairs. The DNA is present inside a viral capsid with a diameter of 40-50 nanometres as well as an icosahedral symmetry (symmetry T=7). This genome is made up of 72 pentameric capsomeres of a specific protein that is VP1, It might also self-assemble it into a compact icosahedron; every and all VP1 pentamer is connected to either VP2 or VP3. A standard polyomavirus genome encodes were between Five or Nine proteins, which are divided into two transcriptional domains depending on when they have been transcribed during infection, known as the early and late regions (Carr et al.,2006).

2.1 Structure of the BK virus:

The BKV-DNA genome is split in three sections.

1. The non-coding regulatory region is one of the most important parts of the genome (NCCR) The NCCR is bidirectional that governs the starting of viral DNA synthesis by controlling transcription of together early as well as late promoters. It includes ori i.e origin of replication. The area between both the start of the Agno gene region, which encodes or start codon that is ATG for the T antigen region, which encodes agnoprotein, (Stoner et al., 2002).

2. In the BKV life cycle, the early section of the BKV genome, which is 2.4 kb long and situated upstream of the origin, is engaged in translation or transcription. Two regulatory proteins are encoded in this region: Alternative splicing produces both antigen that is large tumour (LT-ag) and small tumour (st-ag) antigens from a shared pre-messenger RNA (mRNA) precursor. The early portions of the genome are transcribed in a counter clockwise manner on one DNA strand (Singh et al.,2009).

BKV's late region , which is 2.3 kb in length and situated downstream of the replication origin, is 2.3 kb long. There is three important proteins that is VP1, VP2, and VP3, along with one non-structural agnoprotein. Alternative splicing allows various proteins to be transcribed by producing different mRNA from the same pre-mRNA transcript. Transcriptions in the late area proceed clockwise on the early transcription's

complementary sequence. VP1 comes from a distinct spliced mRNA, while VP2, VP3, and VP4 come from the same mRNA and are read in almost the same gene encoding, but with different start codons (Won et al., 2007).

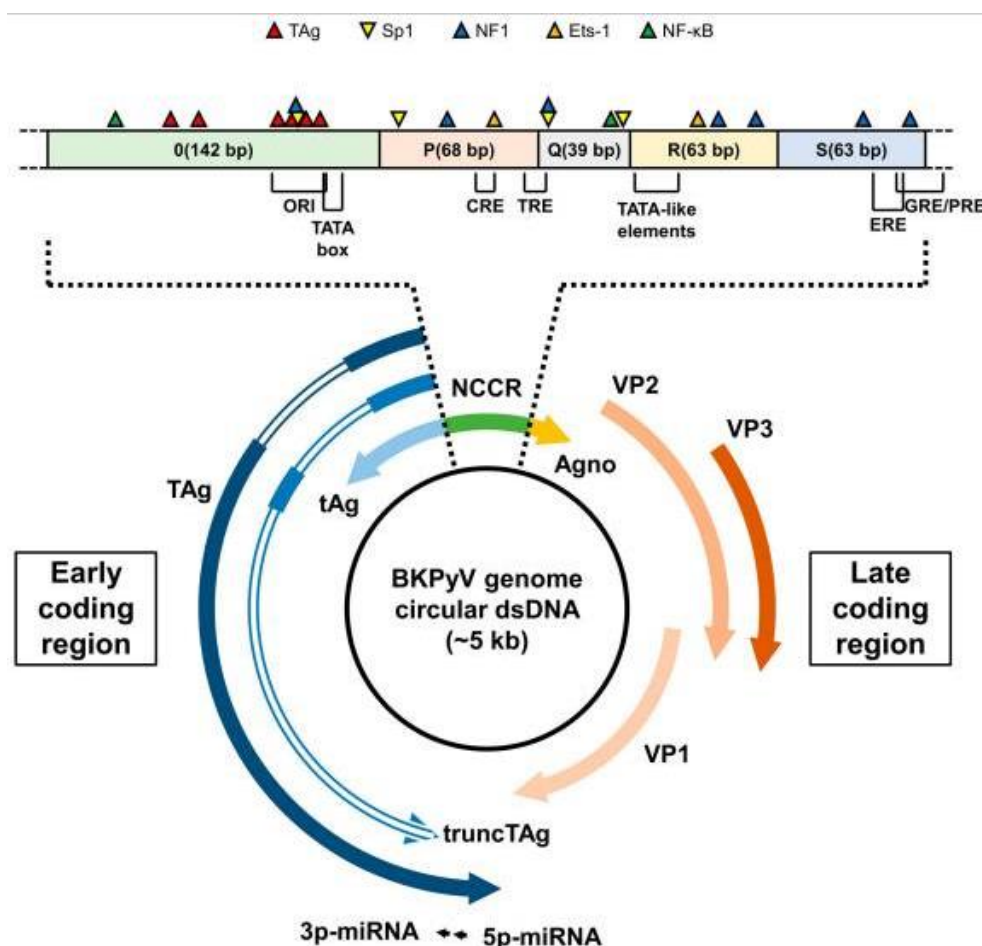


Figure 1-Genomic structure of BKV

2.2 Epidemiology of BKV:

According to population-based BKV prevalence rate studies, 80–90% of children are subjected & infected before the age of 10, with an average age of 4-5 years (Kean et al., 2009). The weakening of BKV immunity after contamination has been indicated by declines in antibody titers over time (Kean et al., 2009; Knowles et al., 2003).

BKV has four unique serotypes as well as subtypes (genotypes): 1,2,3 and 4 with subtype 1 (the most common) found globally, subtype 4 found in East Asia as well as Europe, & subtypes 2 and 3 are very infrequently identified (Liacini et al., 2010).

The method of transmission has also been established by BKV subtyping. Second-generation Japanese-Americans as well as California- Americans revealed that the

European Ib2 lineage predominated in one and other groups, while Ic predominated in Japan, indicating that transmission happens outside of the family (Sharma et al., 2009). There is no obvious link between urine BKV subtype excretion as well as human illness, and the immunological state does not affect BKV subtype excretion (Josephson et al., 2006).

2.3 Clinical features of BKV Infection:

BKV contamination after renal transplant is primarily associated with nephropathy even though BKV infection after stem cell transplant is primarily demonstrated as hemorrhagic cystitis (Rinaldo et al., 2013). BKV-associated hemorrhagic cystitis is seen 25-57 (median value) after stem cell transplantation, while symptoms persist for 10-38 days (median value) (Kloss et al., 2013, Laskin et al., 2013, Megged et al., 2011). Hemorrhagic cystitis of grade 1, 2, 3 and 4 happened as 0%-30%, 16%-57%, 36%-67% and 0%-12% of cases respectively (Kwon et al., 2013, Megged et al., 2011).

BK viruria is identified in 47-52% of all stem cell transplant receivers (Azzi et al., 2005), and only 38%- 44% of them manifest haemorrhagic cystitis (Gorczgnska et al., 2005).

There are at high risk of -

- Acute GVHD
- Old age
- An unrelated donor providing stem cells.
- Allogenic transplantation
- An unrelated donor providing stem cells.

2.4 Haemorrhagic cystitis in stem cell transplant:

It is tiny or gross hematuria associated with lower urinary tract indications like frequent urination, dysuria, urgency & suprapubic pain. Additional etiology like a fungal or bacterial infection, vaginal bleeding & urinary tract mass, or calculi should be rejected (Leung et al., 2005).

According to Bedi et al., haematuria is graded as follows:

- Grade 1- Macroscopic haematuria.
- Grade-2-Microscopic haematuria.
- Grade-3- Clots along with Macroscopic haematuria due to urinary obstruction .
- Grade-4-Macroscopic haematuria with big clots.

Hemorrhagic cystitis in the Bone Marrow Transplant patients is separated into early beginning or before engraftment hemorrhagic cystitis and Late-onset or post-

engraftment hemorrhagic cystitis. Early-onset hemorrhagic cystitis is due to chemotherapeutic agents like cyclophosphamide, ifosfamide, etoposide & busulphan. Early-onset hemorrhagic cystitis is also associated with irradiation provided to the pelvic area during pre-transplant situations and sustained thrombocytopenia before engraftment (Binet et al., 2005). While sustained thrombocytopenia and coagulopathy are associated with hemorrhagic cystitis in the late period after stem cell transplant (Bogdanovic et al., 2004, Kloos et al., 2013, Cesaro et al., 2008).

Some viruses like BKV, JCV, adenovirus, and cytomegalovirus are also reported as a cause of late-onset hemorrhagic cystitis in bone marrow transplant recipients (Bogdanovic et al., 2004, Cesaros et al., 2008).

35%-100% of Bone marrow transplant patients with hemorrhagic cystitis manifest urinary BKV (Bogdanovic et al., 2004, Kloss et al., 2013, Cheuk et al., 2011) compared to 10%-15% in adenovirus infection (Hale et al., 2003), 4%-26% in CMV infection (Haines et al., 2011), and 4-7 % in JC virus infection (Kloss et al., 2013). BKV remains to be the most important pathogen in late onset hemorrhagic cystitis immediately after bone marrow transplantation.

2.5 BKV-related HC epidemiology & risk factors:

BKV is a worldwide distributed virus that affects the massive people. BK-virus infection is endemic and strictly anthroponomic. The virus affects irrespective of gender both men and women, is universal, non-seasonal, and unrelated to the affected population's socioeconomic level. Early childhood infection is most common, with prevalence ranging from 70 to 90 percent in children whose aged lies between 5 to 9 years. Unfortunately, data on BKV incidence is scarce across the globe, particularly in Asia, the Middle East, and Africa. A prevalence of 41.8 percent was found in previous research from Iran. However, recent research found that 80-90 percent of people had seropositivity. The vast majority of initial illnesses are asymptomatic or have very minor symptoms (Krajewski et al., 2020, Najafabadi et al., 2020).

2.6 BKV-related HC pathogenesis:

BK viremia may occur in a broad range of immunodeficiencies, although bone marrow transplant patients (BMT) or renal transplant patients are more prone to this infection.

BKV replication or reactivation are triggered by immunosuppression. While BKV replication is minimal, the patient is symptomless; however, when replication is high, substantial uroepithelial cell lysis occurs, resulting in haematuria. In early

investigations, the BK virus was linked to the development of haemorrhagic cystitis after BMT. Nowadays Hemorrhagic cystitis in BMT patients is common source of morbidity, in some cases it causes death in BMT patients. Clinical indicators range from microscopic haematuria to severe bladder bleeding resulting in renal failure. Bone marrow transplants are performed at a rate ranging from 7% to 68 percent of the time (Addis et al., 2007).

In the improvement of BKV-associated HC amongst HSCT patients, three different stages have been suggested. To begin with, several components of the training regimen induce direct damage to the bladder mucosa. The repetition of latent BKV in urothelial cells is then activated, which is aided by the continuing immunosuppression. The increased excretion of BKV in urine reflects this. Lastly, immunological reconstitution after engraftment triggers immune-mediated urothelial damage, which leads to HC.

3. DIAGNOSIS

The diagnosis of the BK virus has depended strongly on histological methods in post kidney transplant patients. Lately, non-invasive, molecular assays have superseded the diagnostic algorithm in all transplant patients, including HSCT. A sensitive, as well as quantitative PCR assay, can assist in the analytical & therapeutic administration of BKV infection (Azzi et al., 1999 , Erard V 2005).

3.1 Blood

In allogeneic HSCT patients there is slightly higher rate of BKV-HC .they had plasma virus levels of more than 10^3 -4 copies/ml. A negative plasma viral load doesn't indicate the possibility of BKV-HC infection (Atilla et al., 2020).

3.2 Urine

The laboratory diagnosis or investigations of BK virus in haemorrhagic cystitis comprises the identification of decoy cells in cytology and determination of viral load (BK virus) in urine using real-time PCR. In urine the BKV load is $>10^7$ copies/ml showed a susceptibility of <86 percent and a specificity of <60 percent in a paediatric assessment, while in a blood BKV burden $>10^3$ copies/ml had a higher sensitivity & a specificity of almost 86 percent for HC, respectively (Atilla et al., 2020).

3.3 Urine mRNA

The primary limitation of techniques based on viral DNA detection is their inability to distinguish between dormant disease and reactivation. Real-time PCR to analyze mRNA rather than DNA has been reported as an efficient and non-invasive technique for making a presumptive diagnosis of BKV -Nephropathy in mature kidney transplant receivers. Measurement of BKV mRNA levels in plasma or urine has been under study as a highly specific and sensitive method for detection of active BKV replication (Ambalathingal et al., 2017).

Some studies have recommended applying urinary mRNA load of VP-1 to verify active viral replication (Ding et al., 2002). While quantification of BKV DNA in plasma and urine It will be utilized as an important analytical tool and it will be identified in urine sample in the range of 15 to 30% of recipients after renal transplantation throughout the first post-transplantation year (Hussain et al., 2002, Brennan et al., 2005). There is a scarcity of data on the significance of BK m-RNA in HSCT Patients.

3.4 Treatment of BKV-HC:

Hemorrhagic cystitis is generally managed by supportive therapy including hydration, analgesics, platelet transfusion, and bladder irrigation. No definitive therapy for BK is available, though drugs like cidofovir, leflunomide, levofloxacin, and vidarabine have been tried. More BKV-HC treatments aim to heal and regenerate urothelial mucosa by hyperbaric O₂ treatment or topical fibrin sealant. In severe situations, urological intervention including cauterization, embolization, or cystectomy may be required. Immunosuppression reduction has been utilized effectively in kidney transplant patients to avoid as well as cure BKV-related nephropathy but has not proven therapeutic efficacy in HSCT recipients with BK-associated hemorrhagic cystitis

4. RESEARCH GAP

There are few studies from India on BK Virus, but they are mostly on kidney transplant recipients (Bagchi et al., 2017, Singh et al., 2016, Thakur et al., 2011). In India, no prospective study has been done to assess BK virus reactivation and disease burden in stem cell transplants. A publication from India (Arora et al., 2017), has stated a reason for BK-associated hemorrhagic cystitis in an allogenic HSCT recipient, treated with cidofovir.

5. RATIONALE OF STUDY

BKV-HC is one of the most frequent problems after HSCT. It has a high rate of death and morbidity. There seems to be no standard or authorized therapy for BKV-HC, and it is done according to standard operating protocols, which vary identify the causes and severity. Treatment for BKV-HC is lacking in terms of efficient & scientific proof therapies. For the day-to-day treatment of HSCT patients, understanding their origin or characteristics associated with severe illness is critical. Limited data on BK virus reactivation and the frequency of haemorrhagic cystitis in hematopoietic stem cell transplant (HSCT) patients is available from India. This will be pilot research to see whether BK polyomavirus reactivation can be detected in HSCT patients and if there is a clinical connection in our setting. This research will look at how often BK reactivation is among HSCT patients in our clinic. The researchers will also look at the relationship between BK and HC viremia and viruria as well as the advancement of late-onset haemorrhagic cystitis in HSCT recipients.

6. AIM & OBJECTIVES

AIM

- To identify BK polyomavirus and assess its reactivation in HSCT recipients
- OBJECTIVES To identify BK virus DNA in urine & plasma samples in HSCT recipients by real-time quantitative PCR To identify BK virus mRNA in urine & plasma samples in HSCT recipients by real-time quantitative PCR To compare BK virus DNA and BK virus mRNA real-time PCR results in HSCT recipients. To correlate the incidence of haemorrhagic cystitis in HSCT recipients with BK polyomavirus reactivation with respect to risk factors.

7. METHODOLOGY

- **Study design** - Prospective study
- **Sample size** - 100 consecutive HSCT patients

(Sample size prediction based on individual virus could not be done, due to limitation of the number of transplants carried out in practice and the need for screening and testing for long term follow-up)

Inclusion Criteria:

Patients undergoing hematopoietic stem cell transplant.

Exclusion Criteria

1. Patients who refuse to sign a written informed consent form
2. Inability or refusal to follow-up

Study period - for three (3) years

(Patient enrolment is expected to take two years, with a one-year follow-up)

- Study site - Virology Laboratory, Department of Microbiology, AIIMS
- Sampling strategy – After written informed permission, blood and urine samples will be taken.
- **Urine sample** - once a week for the first three months, then at six months and one year.
- **Blood samples** - once a week for the first three months, then at six months and one year afterward.
- Engraftment time (about 2 weeks after HSCT) & biochemical parameters would be recorded.
- All patients who are enrolled will be monitored for a year. A plasma/urine sample will be obtained for testing if indicated or the patient develops haemorrhagic cystitis.
- During the first month after transplantation, particularly until engraftment, patients are admitted to BMT facilities. They are followed up in an outpatient environment after discharge. Family and financial obligations may influence the frequency of OPD follow-up for certain patients, particularly those who live outside of Delhi. All registered patients will have their samples collected regularly, with the patient's convenience in mind.
- The research participant's plasma/ urine samples will be collected every week for the first three months, then at six months and one year afterward. The study participants would be monitored for a year and screened for possible viral reactivation and illness.

7.1 Methods:

- BK polyomavirus testing would be done for screening as well as for suspected clinical illness evaluation.
- Post-transplant assessment- BK polyomaviruses will be tested in the patient's sample collected weekly for up to three months (weekly till 3m), at 6 months and 1 year.

- Patients will be classified under two types of risk either as high or low. Depending on type of transplant, pre-transplant screening, and immunosuppressive medications.
- BK polyomavirus will be tested weekly in the urine/plasma using quantitative real-time PCR.

	Virus	Method	Sample	Frequency
Post-transplant (Recipient)	BK polyomavirus	Real-time PCR	Blood, Urine	Weekly till 3 months, then at 6 months and 1 year

Table-1 Methodology

- Following the transplant, the person's blood counts will be documented, as well as the day of engraftment. Engraftment is described as a neutrophil amount of more than 500/l and a platelet count of < 20,000/l two weeks after HSCT.
- Clinical illness evaluation - Patients will be carefully monitored for one year after HSCT for any signs of viral infection and checked as needed using standard laboratory molecular techniques.
- Plasma and urine samples will be examined for BK detection in enrolled individuals with probable haemorrhagic cystitis or viral reactivation.

Virus	Clinical condition
BK polyomavirus	Hemorrhagic cystitis, GVHD Rarely – nephropathy, ureteric stenosis

Table-2 Clinical Diagnosis

7.2 Molecular testing – BK Polyomavirus:

Urine Samples:

3-5 ml urine sample collected in a sterile container. Samples will be aliquoted and stored at – 80°C for further processing.

Plasma:

A total of 3-5 ml whole blood sample will be collected aseptically by peripheral venipuncture. The sample will be collected in an EDTA vacutainer. Plasma will be separated, aliquoted, and stored at – 80°C for further processing.

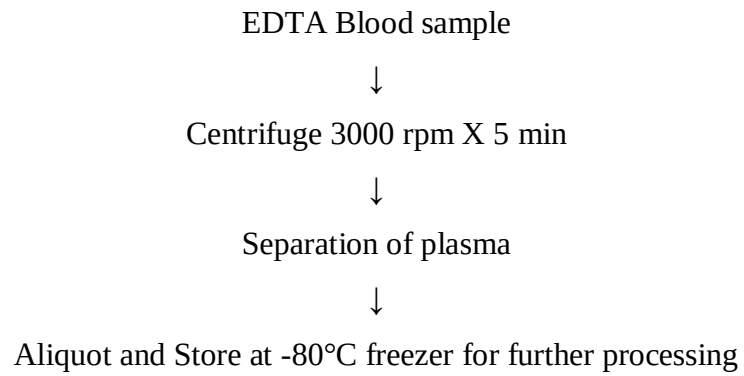


Figure-2 Sample Processing Flow Chart

7.3 Molecular Analysis:

All samples collected from patients – urine and plasma will be screened for BK virus by Real-time PCR. The following procedures will be followed for each sample.

1. DNA Extraction
2. BK Real-time PCR
3. BK m-RNA Real-Time PCR

Workflow:

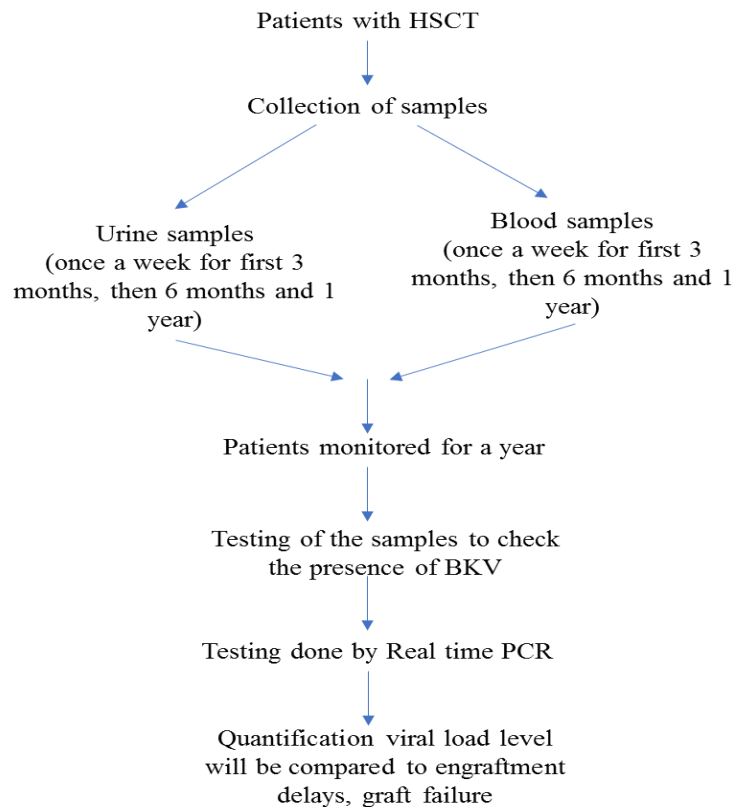


Figure-3 Work Plan

DNA Extraction:

The DNA extraction will be done by using commercial Qiagen QIAamp Blood- DNA mini kit for plasma & Qiagen QIAamp Viral RNA mini kit from a urine sample. The eluted DNA will be stored at -20°C freezer.

Real-Time PCR:

- Real-time PCR assay will be accomplished using Gene Proof BK/ JC Virus Real-time PCR Kit (Gene Proof, Brno, Czech Republic), as per manufacturer's instructions on Agilent Aria Mx real-time PCR (Agilent Technologies) instrument.
- In a single reaction tube, the PCR test identifies & differentiates BK as well as JC viruses. The ready-to-use master mix comprises Uracil-DNA glycosylase (UNG) as well as dUTP, which eliminates the possibility of amplification product contamination.
- The real-time quantitative PCR targeted a particular conservative DNA sequence that overlaps the gene border for the VP1 as well as VP2 proteins.

7.4 Molecular testing – BK mRNA:

RT-PCR:

- RT-PCR assay will be accomplished by utilizing published primers and probe for BK m-RNA Real-time PCR (Ding et al., 2002)
- The sequences, as well as the site of the primers and the probe, are observed as Sense-TGCTGATATTTGTGGCCTGTTTACTA (2355-2380); Anti Sense-CTCAGGCGGATCTTAAAATATCTTG (2438-2414); Probe AGCTCTGGAACACAACAGTGGAGAGGCC (2383-2410)
- The real-time quantitative PCR detects the conservative BKV VP1m-RNA. The primers and probe complement a specific region of VP1 gene that is extremely protected throughout the main subtypes of BKV (GenBank accession number J02038) but mismatches the JCV VP1 region.

8. DATA COLLECTION AND ANALYSIS

To analyse any delays in transplant or treatment outcomes, thorough clinical data and lab data will be recorded. The study's sample collection, testing, and analysis will all be done in real-time. Microsoft Excel will be used to input all of the data. SPSS software and suitable statistical tools will be used to analyse the data.

The risk of viral reactivation and illness will be evaluated. Quantification/ viral load level will be compared to engraftment delays, graft failure, and poor short- and long-term outcomes. Using ROC curves, an attempt will be made to create a plasma viral load cut-off for possible correlation with a viral illness, which will be compared to cut-offs suggested in worldwide analysis and recommendations.

We will look at whether the presence of the virus as a primary, reactivation, or reinfection affects both clinical outcome and illness progression.

9. POSSIBLE OUTCOMES AND LIMITATIONS

The goal of the research is to learn more about the epidemiology of BK polyomavirus reactivation in HSCT patients and how it relates to clinical outcomes. It will offer information on the need for polyomavirus monitoring or screening for stem cell transplant patients in the Indian healthcare system. For Indian HSCT patients, the research will try to establish a significant viral load cut-off for BK polyomavirus in urine and plasma.

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PATIENT/ PARENT/ LAR PARTICIPANT INFORMATION SHEET

Title of the Study: “BK Polyomavirus identification in hematopoietic stem cell transplant recipients”

Dear Participant

Thank you for your interest in this study. Please read the following information carefully and clarify your doubts with us, if any.

Introduction and purpose of the study:

It has been seen that some viral infections can cause diseases or graft related problems in patients receiving bone marrow transplant. Some viruses cause infection after transplant and some viruses which are generally harmless in healthy people can reactivate in patients due to low immunity. While some of these reactivated viruses cause no to minimal effect, some can have diseases and severe consequences. The viral infections in these patients can be detected by testing blood and urine for presence of virus. The study will help in identifying presence of one of such viral infection, BK polyomavirus in post-transplant patients. This information aims to benefit medical community and patients in an effort to understand and ultimately prevent diseases from these infections in transplant setting.

What do I have to do in this study?

If you agree to join this study, you will agree to these things:

- Medical information about you/ your child/ ward will be obtained from you.
- Blood and urine samples will be collected from you/ your child/ ward for testing of viral infections after enrolment. The samples will be collected with your weekly follow-up, up to 3 months, then at 6 month and 1 year. About a teaspoon (3 ml) of blood will be collected from you/ your child/ ward at the time of collecting blood for routine laboratory diagnosis. You will also be requested to collect about one teaspoon (5 ml) of urine sample from yourself/ your child/ ward in the container provided with instructions about how to collect.
- The sample from you/ your child/ ward will be stored for research analysis related to the current project at a later time.

What are the risks in taking part in this study?

There is no additional risk in this study. Minor bleeding can occur from the site of puncture and there can be mild temporary pain. However, this is a safe procedure and routinely performed in the hospital by trained personnel only. Adverse events related to blood collection, if any, will be managed at AIIMS.

What are the benefits?

There is no direct benefit to you/ your child/ ward for participating in this study, but your cooperation may help to improve the knowledge of viral infections in post-transplant patients. This work will provide insight into the disease burden of polyomavirus and other viral infections in transplant patients. It will help your treating doctor in understanding the implications of viral infections in causing disease and graft related problems.

Maintenance of confidentiality of records: Your personal information including your name, address, contact details, profile/designation, institute name, etc. will be kept confidential during and also after the completion of the study. The research findings will be shared through publications and conferences without disclosing your identity.

Freedom of individuals to participate: Your participation in the study is voluntary. You can choose to say No and this will involve no penalty. You may withdraw at any time during the study when you wish to. This will not affect your treatment process in the hospital.

Problems or questions: If you have any questions about the study you may contact us with the contact details provided to you. You will also be asked to provide your home address and contact details in case the study organizers need to contact you. If you feel that your questions have not been answered to your satisfaction or you are being forced to participate in this study, then you can contact the higher authority (Dr. Megha Brijwal), and report it.

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PATIENT/PARENT/LAR INFORMED CONSENT FORM

Protocol/Study Number: _____

Participant identification number for this trial: _____

Title of Project **“BK polyomavirus identification in hematopoietic stem cell transplant recipients”**

Name of Principal Investigator: Dr. Megha Brijwal

Tel. no. 9868994329

The Content of the information sheet dated _____, that was provided have been read carefully by me/explained in details to me, in a language that I comprehend, and I have fully understood the contents, I confirm that I have had the opportunity to ask questions.

The nature and purpose of the study and its potential risks/benefits and expected duration of the study, and other relevant details of the study have been explained to me in details. I understand that my/my child's/ward's participation is voluntary and that I am free to withdraw consent at any time, without giving any reason, without my /my child's/Ward's medical care or legal right being affected.

I understand that the information collected about me/my child/ward for this research and sections of any of my/my child's/ward's medical notes may be looked at by responsible individuals from participating institutions. I give permission for these individuals to have access to my/my Child's/Ward's records.

I agree for the participation in the above study.

(Signature/Left Thumb Impression)
Patient/Parent/LAR

Date:
Place:

Name of the Participants: _____
Son/Daughter of: _____
Complete Postal address: _____

This is to certify that the above consent has been obtained in my presence.

Signature of the Principal Investigator

Date:
Place:

1. Witness-1

2. Witness-2

Signatures

Signatures

Name:
Address:

Name:
Address: